

GIESSEN, Germany

WMO: 105320

Lat: 50.6017N

Lon: 8.6439E

Elev: 669

StdP: 14.34

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.5	15.4	5.0	7.3	13.6	9.6	9.2	17.6	23.6	43.3	20.4	42.4	5.2	20	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	85.3	66.5	82.1	64.8	79.0	63.4	68.4	80.1	66.6	78.1	65.0	75.6	7.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	92.7	71.8	62.6	87.1	70.7	61.0	82.2	69.2	33.0	80.5	31.5	78.4	30.3	75.8	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.5	13.6	9.3	91.4	7.9	2.3	3.7	93.0	-1.0	94.4	-5.4	95.7	-11.1	97.3		
(5)			8.9	72.1	7.9	1.3	3.2	73.0	-1.5	73.7	-5.9	74.4	-11.7	75.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	33.4	33.2	41.3	47.1	55.9	61.3	65.6	65.0	58.2	50.0	40.2	36.3		
	DBStd	13.36	9.35	8.20	6.27	6.27	6.47	6.30	5.69	5.41	5.28	6.08	6.53	7.29		
	HDD50	2209	514	470	276	128	19	1	0	0	2	76	296	427		
	HDD65	6016	979	889	736	536	286	145	64	68	214	465	743	891		
	CDD50	1869	0	1	5	41	203	341	483	466	247	76	4	2		
	CDD65	199	0	0	0	0	5	35	82	69	8	0	0	0		
	CDH74	2076	0	0	0	2	109	348	845	682	88	2	0	0		
	CDH80	513	0	0	0	0	7	64	240	192	10	0	0	0		
	WSAvg	5.8	7.0	6.2	6.6	6.2	5.6	5.6	5.4	5.1	5.0	5.3	5.7	5.9		
	PrecAvg	28.9	2.5	2.0	1.9	1.8	2.4	2.6	3.0	2.5	2.3	2.4	2.6	2.8		
(15) Precipitation	PrecMax	36.6	5.3	5.2	4.1	3.7	4.6	5.3	5.9	5.7	5.3	7.2	4.2	7.5		
	PrecMin	22.6	0.2	0.2	0.4	0.1	0.6	1.1	1.3	0.2	0.9	0.4	0.2	0.7		
	PrecStd	3.5	1.2	1.2	1.0	0.8	1.1	1.1	1.1	1.2	1.0	1.4	0.9	1.3		
	PrecStd	3.5	1.2	1.2	1.0	0.8	1.1	1.1	1.1	1.2	1.0	1.4	0.9	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	57.0	65.3	72.4	80.8	85.6	89.3	90.5	81.5	71.1	57.2	56.3		
		MCWB	48.7	50.3	51.9	55.4	61.4	67.5	66.9	66.9	62.9	59.2	52.5	51.3		
	2%	DB	49.3	51.5	58.4	67.3	76.9	81.6	85.9	85.2	76.4	65.8	53.8	51.5		
		MCWB	45.7	47.1	48.8	53.0	59.2	64.7	66.9	65.9	62.6	57.6	50.5	48.4		
	5%	DB	47.1	47.2	53.9	63.6	73.9	78.1	82.5	81.1	72.6	61.8	51.4	49.0		
		MCWB	44.3	43.3	46.8	51.3	58.2	62.7	65.5	64.4	61.2	55.9	48.4	46.7		
	10%	DB	44.8	43.7	50.9	59.2	69.9	74.5	79.1	77.3	68.3	59.0	49.1	46.5		
		MCWB	42.4	40.5	45.8	48.8	56.8	61.7	64.0	62.8	59.0	53.9	46.5	44.5		
	0.4%	WB	49.1	50.9	53.5	56.7	63.7	70.0	70.9	70.0	66.6	61.5	53.3	51.3		
		MCDB	51.9	55.9	60.3	69.7	74.6	80.3	84.0	84.2	75.5	67.9	54.6	54.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.7	47.4	50.2	54.6	61.3	67.2	68.6	67.5	64.1	58.8	51.2	49.0		
		MCDB	49.0	50.6	56.6	64.2	72.8	77.5	81.1	79.1	72.2	63.6	53.4	51.3		
	5%	WB	44.6	43.8	48.2	52.2	59.5	64.8	66.9	65.9	62.5	56.7	48.8	46.7		
		MCDB	46.5	46.6	53.2	61.1	70.6	74.4	79.3	77.2	70.0	60.8	50.9	48.6		
	10%	WB	42.4	40.9	46.0	50.2	57.7	62.8	65.1	64.4	60.6	54.8	46.8	44.6		
		MCDB	44.8	43.3	50.5	58.2	67.8	71.8	76.3	74.7	66.7	58.3	48.7	46.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	11.3	13.6	16.7	18.7	17.3	19.2	19.2	16.7	13.5	9.2	7.6		
		MCDBR	9.4	15.0	20.2	24.4	26.3	24.5	25.8	26.4	23.9	17.6	10.7	10.2		
		MCWBR	8.2	10.9	12.2	12.4	12.2	10.7	10.9	10.5	11.8	10.4	8.4	8.8		
		MCDBR	9.3	14.2	17.6	21.5	22.9	22.0	22.6	21.8	20.0	15.6	10.0	9.8		
	5% WB	MCWBR	8.6	11.2	11.9	11.9	11.5	10.6	10.7	9.6	11.1	10.0	8.4	8.6		
		MCWBR	8.6	11.2	11.9	11.9	11.5	10.6	10.7	9.6	11.1	10.0	8.4	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.340	0.375	0.402	0.406	0.407	0.415	0.410	0.379	0.364	0.339	0.312		
	taud		2.400	2.341	2.276	2.251	2.268	2.283	2.258	2.300	2.370	2.394	2.397	2.393		
	Ebn at Noon		227	249	260	266	270	270	265	260	256	237	213	208		
	Edh at Noon		20	27	34	39	41	40	41	37	31	25	20	17		
(44) All-Sky Solar Radiation	RadAvg		238	460	815	1301	1535	1701	1610	1386	995	582	263	176		
	RadStd		35	69	116	163	168	160	198	127	138	62	36	19		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page